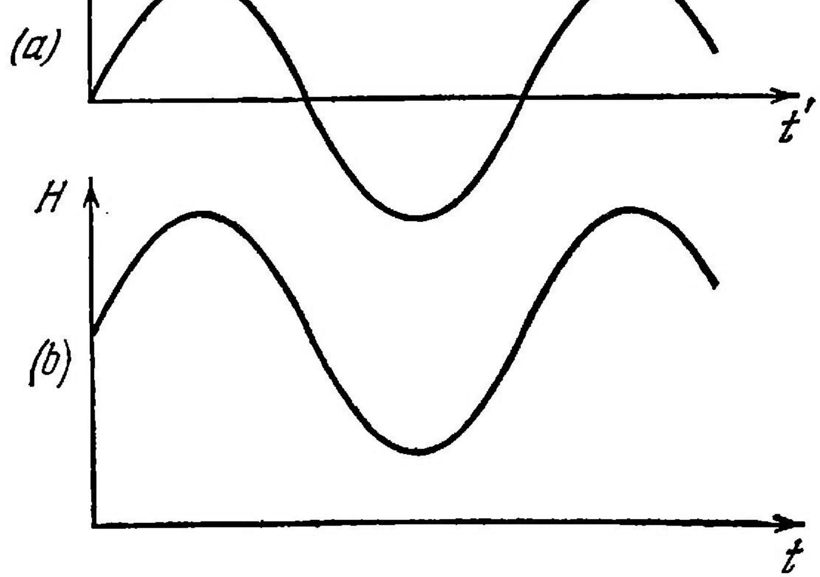

Fig. 481

Hence,

$$\omega = 3\times 10^{10}\sqrt{\frac{ld}{N^2 A_1 A_2}} = 3\times 10^{6}\ \frac{1}{\text{second}}$$

643. The frequency of natural oscillations of a circuit is determined from the Thomson formula

$$\omega = \frac{1}{\sqrt{LC}}$$

(a) If the coil contains the copper core, the periodic changes of the magnetic field of the coil will produce in the core eddy currents whose magnetic field will weaken the magnetic field of the coil. This will reduce the inductance of the coil and, consequently, increase the frequency ω.

(b) If the ferrite core is moved into the coil, the magnetic field of the latter will increase. Accordingly, the inductance L of the coil will grow and the frequency ω will decrease.

644. Undamped oscillations will appear in the system (if the small losses of energy for the radiation of electromagnetic waves are neglected.) When the charge is distributed equally between the capacitors, the energy of the electrostatic field is minimum, but the current intensity and the energy of the magnetic field are maximum. The total energy does not change, but one kind of energy is converted into another.

645. The displacement of the electron beam by the voltage supplied along the vertical can be written as

$$x = \frac{lL}{2dV} V_{01} \cos \omega t = a \cos \omega t$$